

Particulate Matter Monitoring & Health Effects

Issue 2026-08-28 | Entry window 28 August 2026 | 19 records screened in scope

19RECORDS IN SCOPE
(54 EXCLUDED)**8**SUBTOPIC CLUSTERS
OCCUPIED (OF 10)**8**RECORDS WITH NO
HUMAN ENDPOINT**4**EFFECT ESTIMATES
FROM 2 STUDIES**6**TIER-A
RECORDS

Signal of the day

Sulfate carries three-quarters of the $PM_{2.5}$ mixture effect on infarct mortality — and, separately, coating chemistry moves black carbon's mass absorption cross section by $\sim 1 \text{ m}^2/\text{g}$ overnight. Both results penalise mass-only instruments.

Li et al. (*Environ Sci Technol Lett*) fit a causal mixture model to county-level US acute myocardial infarction mortality against $PM_{2.5}$ and its EC / OC / SO_4^{2-} / NO_3^- constituents in a difference-in-differences design. Total mass gives **RR 1.019 (95% CI 1.013–1.025) per IQR**; the constituent mixture gives **RR 1.116 (1.101–1.130) per quartile**, with SO_4^{2-} **weighted 0.762** of that total and an east–west split (SO_4^{2-} 0.707 in the east, OC 0.485 in the west). The mass estimate is not wrong, it is simply a diluted average over a mixture whose components are not interchangeable. Zhang et al. (*ACP*) supply the instrumentation-side counterpart from Shenzhen: SP2 + SPAMS single-particle measurement shows nighttime heterogeneous and daytime photochemical aging proceed by different chemistry (nitrate-first vs. nitrate-then-sulfate) yet converge on nearly the same **MAC enhancement at 532 nm — $\sim 0.8 \text{ m}^2/\text{g}$ overnight, $\sim 1.0 \text{ m}^2/\text{g}$ by day** on a $1.2 \mu\text{g}/\text{m}^3$ rBC background with 155–170 nm core MMD. For a distributed low-cost network this is the operative number: an optical channel calibrated against a fixed mass-scattering assumption absorbs the whole diurnal aging cycle as apparent concentration drift.

What changed today

- **Exposure assessment is the largest cluster (5 records), and for once it is not all epidemiology.** It holds a satellite–surface fusion product, a new summary statistic, a filter-composition campaign, a wave-channel source-process experiment and a bioaerosol survey. The common thread is that all five are about *how the exposure number is constructed*, not what it does to people.
- **A 1 km daily $PM_{2.5}$ surface for Bangladesh now exists.** Cui et al. gap-fill MAIAC AOD — more than 60% of retrievals are missing — with LightGBM, then push the complete AOD through an attention-based TabNet to ground level. Held-out $R^2 = 0.95$, RMSE $17.41 \mu\text{g}/\text{m}^3$ over a national mean of 68–75 $\mu\text{g}/\text{m}^3$. The RMSE is roughly a quarter of the mean; that is defensible at these concentrations but it is not a monitoring-grade product, and the paper's own DHS linkage is ecological.
- **Citizen-science monitoring gets its first proper critical review in this watch.** Marcon et al. (*Curr Environ Health Rep*) is explicit that AI-assisted calibration and distributed sensor networks are the live methodological frontier, and equally explicit that representativeness, not sensor accuracy, is the binding constraint on policy uptake — an ordering worth taking seriously.
- **A threshold-free concentration-frequency metric.** Hopke et al. define the $PM_{2.5}$ Magnitude–Rank Index (largest k with $\geq k$ days at $\geq k \mu\text{g}/\text{m}^3$), applied to 918 US MSA-years. It correlates 0.83–0.96 with the annual mean, the 98th percentile and exceedance counts — which is both the selling point and the problem: a metric that correlates 0.96 with days $\geq 15 \mu\text{g}/\text{m}^3$ has to earn its place on interpretability, not information content.
- **Low-cost sensors as the primary instrument in a health-risk paper.** Wongsasuluk et al. run five DustBoy stations for a year in a Chiang Mai road underpass (385,622 records) and build a stepwise MLR forecast. No co-location against a reference monitor is reported in the abstract, so the risk estimates inherit whatever bias the DustBoy units carry in a high-humidity, high-loading tunnel — exactly the regime where nephelometric error is largest.
- **Policy-scale co-benefit modelling.** Zhang et al. (*Nat Commun*) put hydrogen shipping at $\sim 129,500$ (68,300–202,200) averted premature deaths annually and US\$306 bn (146–488) saved — but **75% of that is ozone**, only 25% $PM_{2.5}$. Read as a PM result it is much weaker than the headline suggests.
- **Empty clusters.** *Respiratory & allergic* and *Reproductive & developmental* are both unoccupied; respiratory was last empty on 23 August.

Corpus at a glance

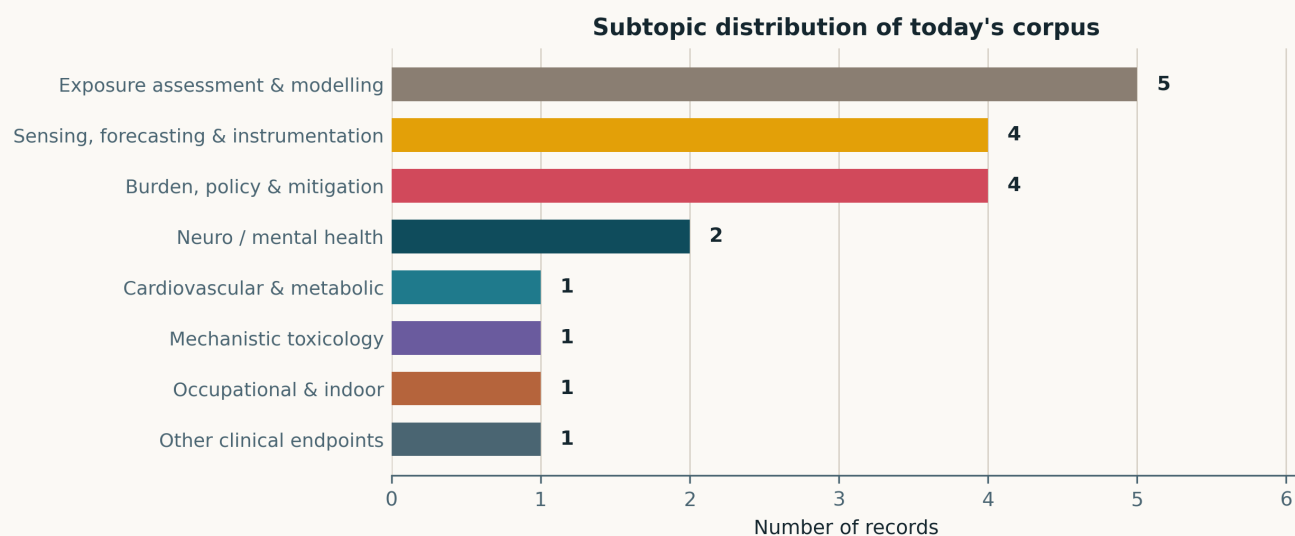


Fig. Records per subtopic cluster, 19 records over **eight of ten** clusters. *Exposure assessment & modelling* leads with 5, followed by *Sensing* and *Burden, policy & mitigation* at 4 each — so **13 of 19 records (68%)** sit on the **measurement/modelling/policy side** rather than the clinical side. *Respiratory & allergic* and *Reproductive & developmental* are empty.

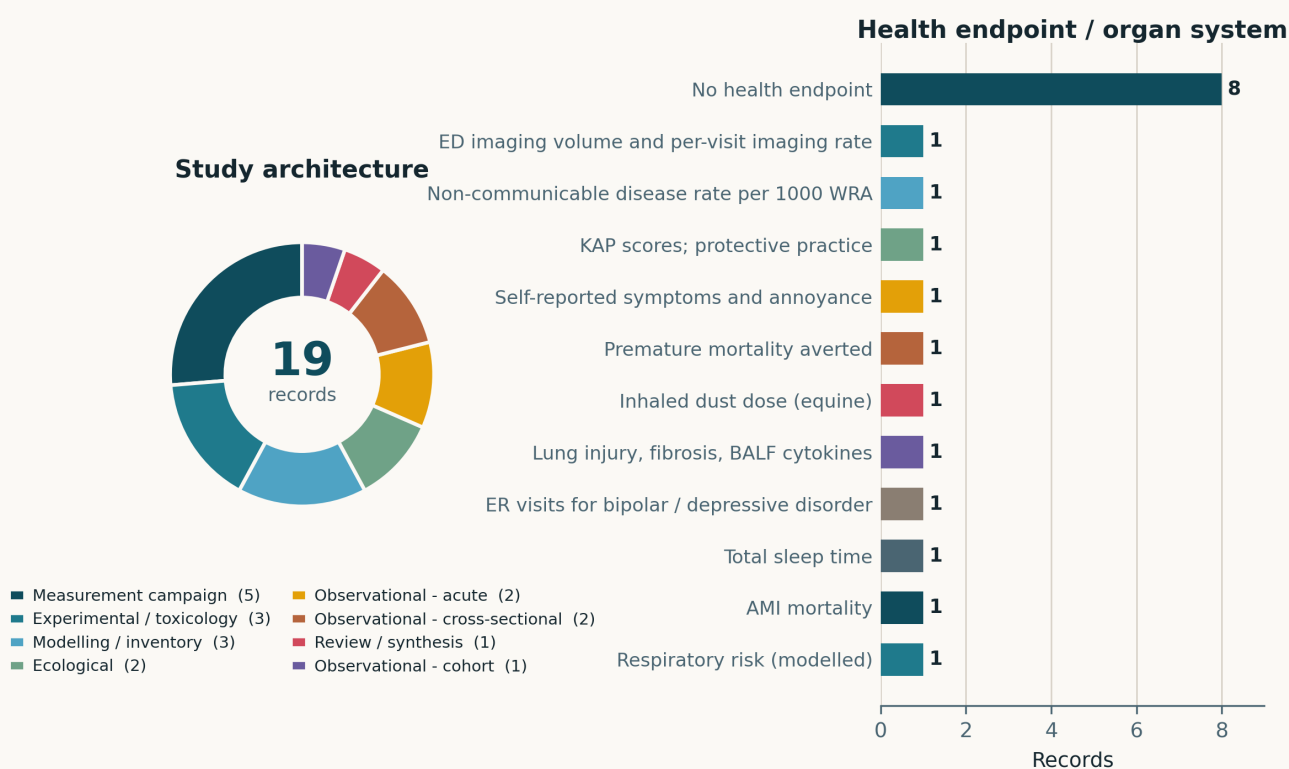


Fig. Left — study architecture. *Measurement campaign* 5, *Experimental / toxicology* 3, *Modelling / inventory* 3, *Ecological* 2, *Observational — acute* 2, *Observational — cross-sectional* 2, *Observational — cohort* 1, *Review / synthesis* 1. The *Experimental* bar is heterogeneous by construction: it pools a murine instillation study with a wave-channel aerosol-generation experiment and a dark-field microscopy method paper, because the grouper has no bench-instrumentation bin. Right — health endpoint. **Eight records carry no human endpoint** — three instrument, three exposure-construction and two source/bioaerosol — so 42% of the corpus is methodological. No endpoint appears more than once.

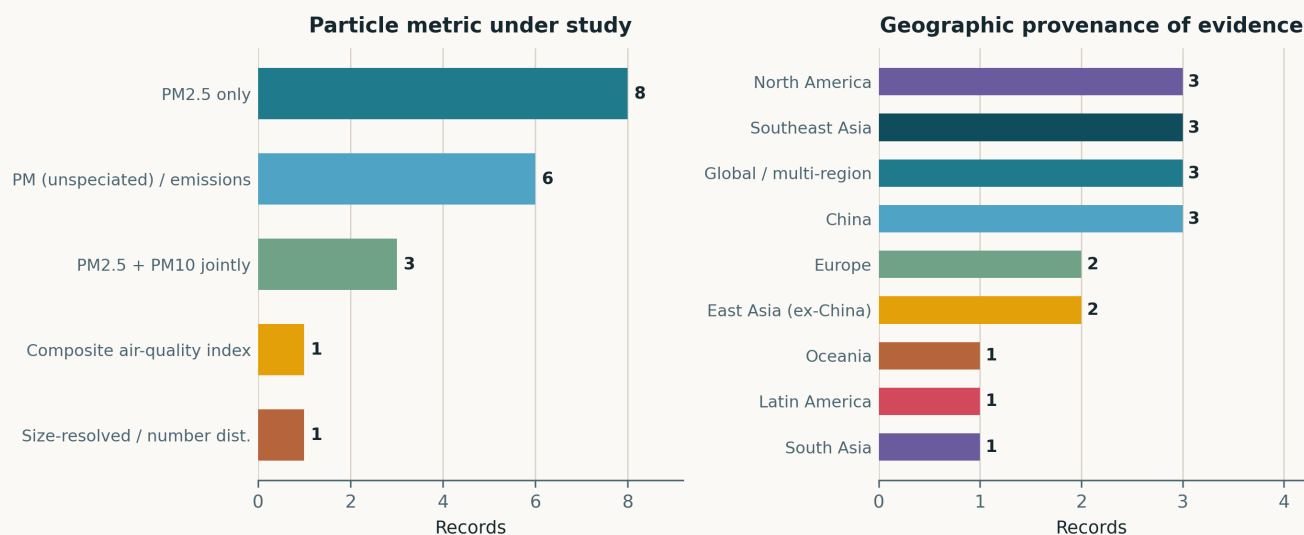


Fig. Left — particle metric. *PM_{2.5} only* 8, *PM (unspeciated) / emissions* 6, *PM_{2.5} + PM₁₀ jointly* 3, and one each for *size-resolved / number distribution* (Leibensperger et al., SSA 0.008–20 μm) and *composite air-quality index*. Note the recurring limitation: both speciation-driven records of the day — Li et al.'s SO_4^{2-} -weighted mixture and Wang et al.'s five-constituent CHARLS analysis — bin as *PM_{2.5} only*, because the grouper still has no speciation bin. Right — geography: China 3, North America 3, Southeast Asia 3, multi-region 3, Europe 2, East Asia ex-China 2, and one each from South Asia, Latin America and Oceania. **Nothing from Africa**; the last African record was 20 August, eight issues ago.

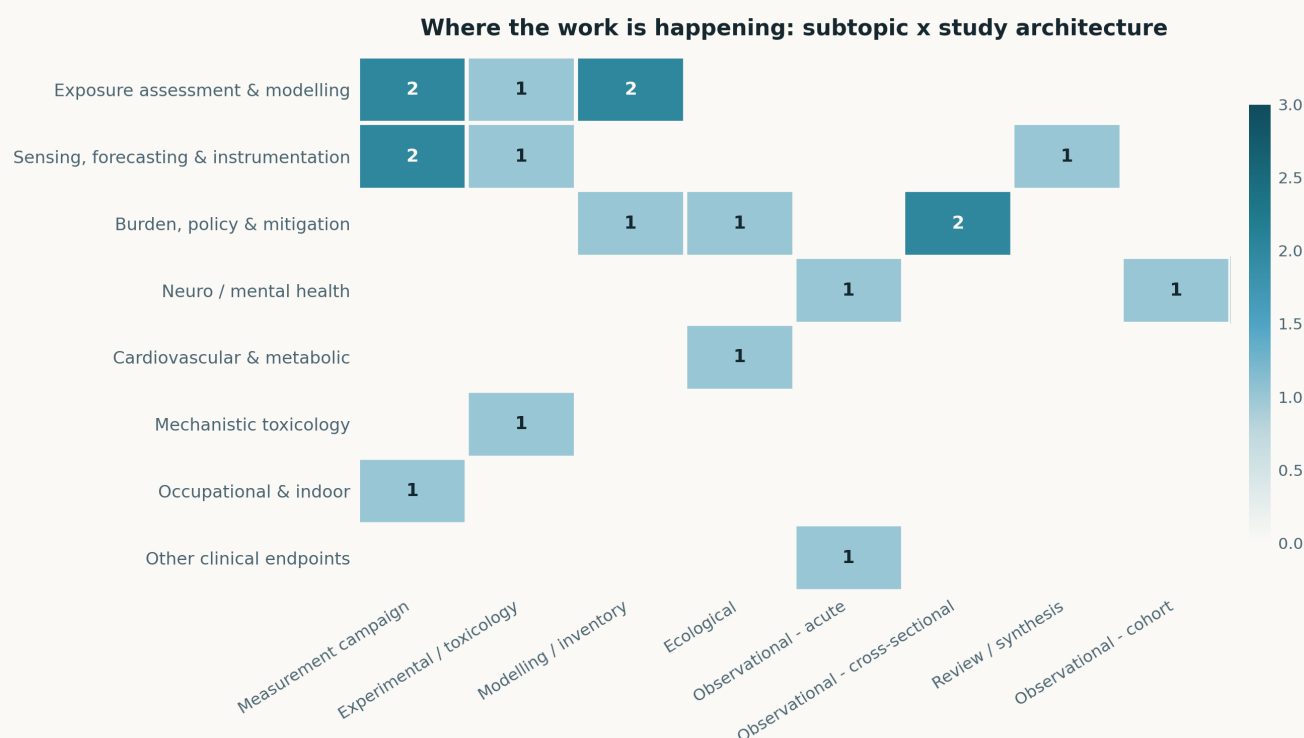


Fig. Subtopic $\times$ architecture cross-tabulation. Nineteen records over eight clusters and eight architectures gives a near-diagonal sparsity map: **four cells hold 2** — *Sensing $\times$ Measurement campaign*, *Exposure assessment $\times$ Measurement campaign*, *Exposure assessment $\times$ Modelling / inventory* and *Burden $\times$ Observational — cross-sectional* — and the remaining eleven occupied cells hold 1. Read as sparsity, not structure.

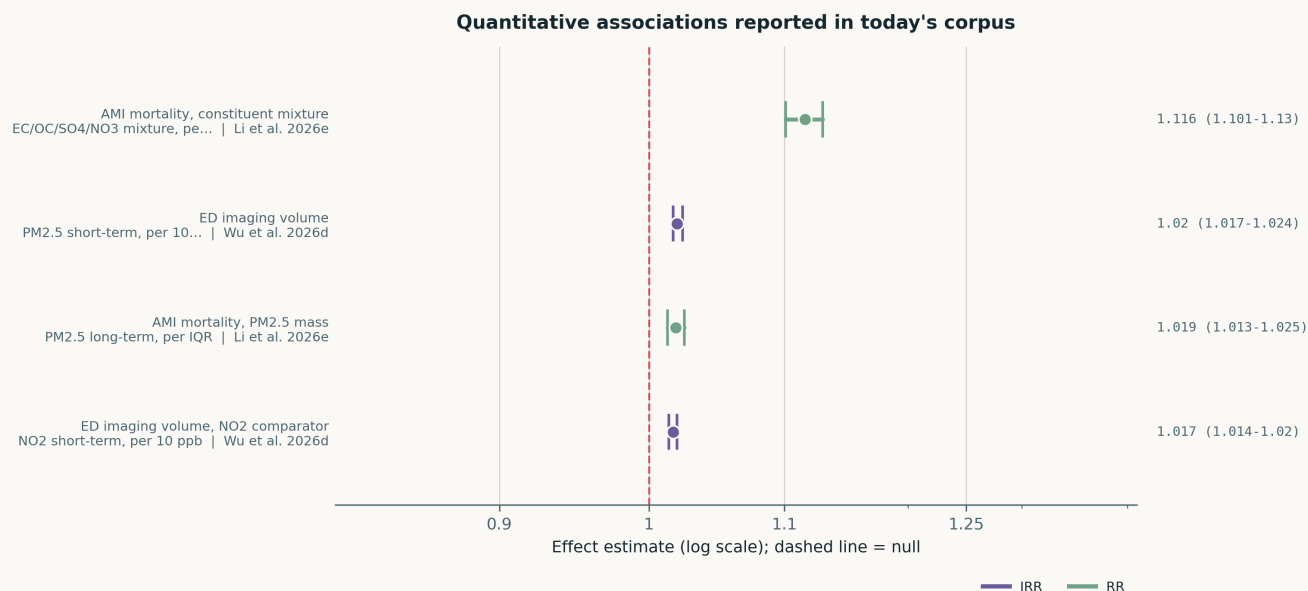


Fig. Four interval-bearing estimates from two studies. The two Li et al. estimates are not on a common contrast — 1.019 is per IQR of total PM_{2.5} mass, 1.116 is per quartile of the four-component constituent mixture — so the visual gap between them is a contrast difference as much as a mixture-versus-mass difference, and should not be read as an effect ratio. The two Wu et al. estimates are mutually comparable in construction (same case-crossover, same Poisson offset structure) but the PM_{2.5} and NO₂ contrasts differ in unit (per 10 µg/m³ vs. per 10 ppb); the NO₂ row is carried only as a within-study comparator. The day's other health results — the CHARLS sleep-duration coefficients and the Taiwanese mood-disorder case-crossover — report on scales that do not belong on a ratio axis and are excluded.

Life-course windows addressed today (records may span several stages)

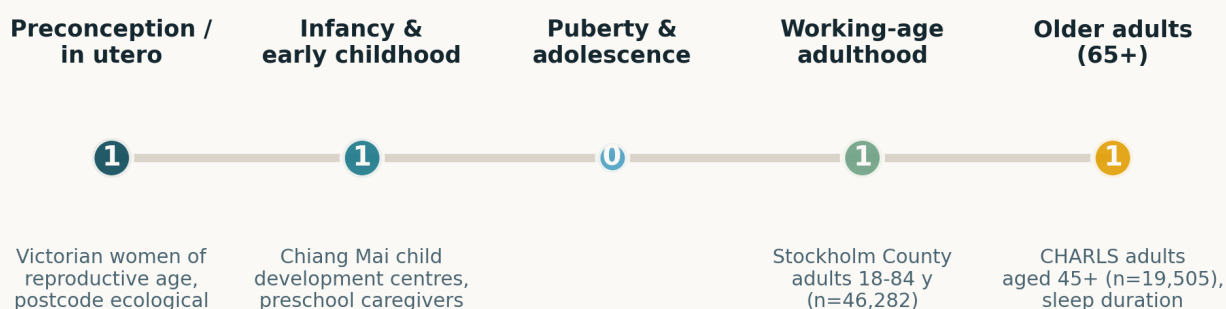


Fig. Life-course windows, counted only where a record's population is explicitly anchored to a stage. Four records qualify across four of five windows; the puberty/adolescence window is empty again. The preconception cell holds the Victorian women-of-reproductive-age ecological linkage, which is a postcode-level correlation study and does not resolve any individual exposure window — it occupies the cell on population definition alone.

Paper-level digest

Ordered by cluster; the 28 August entry window only.

Sensing, forecasting & instrumentation

4 records

Zhang et al. 2026h Atmos Chem Phys

Contrasting nighttime heterogeneous and daytime photochemical aging drive the optical evolution of black carbon

Integrated SP2 and single-particle aerosol mass spectrometry over an urban Shenzhen campaign, mean refractory BC $1.2 \mu\text{g m}^{-3}$, core mass median diameter 155–170 nm. The coating-to-core mass ratio rises continuously through the diurnal cycle by two mechanistically distinct routes — nighttime heterogeneous coating growth with enhanced nitrate signal, and daytime photochemical aging with rapid nitrate accumulation followed by sulfate and oxidised organics. Both converge on similar mass absorption cross section enhancement at 532 nm: $\sim 0.8 \text{ m}^2 \text{ g}^{-1}$ **overnight**, $\sim 1.0 \text{ m}^2 \text{ g}^{-1}$ **by day**. No uncertainty is given on the MAC enhancements and it is a single city over one campaign, so the magnitudes should not be transported. Directly relevant to distributed sensing: any optical channel with a fixed absorption or scattering assumption re-expresses this aging cycle as concentration drift with a diurnal period.

doi: 10.5194/acp-26-12197-2026

Marcon et al. 2026 Curr Environ Health Rep

Citizen science in air pollution monitoring: methodological approaches, effectiveness and integration with health research

Critical review of participatory air-quality monitoring — low-cost sensors, wearables, community co-creation — and its uptake into environmental health research. The review identifies AI-assisted calibration and distributed sensor networks as the methodological frontier and argues that hyperlocal citizen data materially improves spatial resolution and risk communication. Its useful contribution is the ordering of the barriers: **representativeness and policy uptake, not sensor accuracy**, are named as the binding constraints. No quantitative synthesis, no pooled performance figures and no inclusion criteria are reported in the abstract, so it is a framing document rather than an evidence base. For a distributed low-cost network this is the argument that siting design deserves as much engineering attention as the calibration pipeline.

doi: 10.1007/s40572-026-00559-1

Wongsasuluk et al. 2026 Environ Sci Pollut Res Int

PM_{2.5} exposure and respiratory health risk for vulnerable populations in an urban underpass: a predictive MLR approach, Chiang Mai

Five DustBoy low-cost stations along the Don Kaew road tunnel, August 2022 to July 2023, yielding **385,622 records** across PM_{2.5}, PM₁₀, CO, CO₂, NO₂, SO₂, O₃, TVOC, CH₂O and meteorology. A stepwise multiple linear regression stratified by weekday/weekend and day/night gives a high reported goodness of fit, and a comparative risk assessment converts predicted concentrations into inhaled dose and relative risk against Thai and WHO exposure factors. The inference-threatening gap is that **no co-location against a reference monitor is reported**: a semi-enclosed tunnel is a high-humidity, high-loading environment, precisely where nephelometric low-cost sensors over-read most, and every downstream risk number inherits that bias uncorrected.

doi: 10.1007/s11356-026-38171-3

Xiong et al. 2026 Atmos Chem Phys

In situ real-time determination of SO₂ photochemical oxidation in nanoscale sea salt aerosols based on dark-field microscopy

Planar-waveguide dark-field microscopy tracks hygroscopic growth factors of single NaCl nanoparticles under UV irradiation and SO₂, with the Zdanovskii–Stokes–Robinson rule converting growth to composition and hence to reaction rate. Reported first-order rate constant **0.6523 h⁻¹ for 100 nm NaCl**, with a **non-monotonic size dependence** — rising from 50 to 200 nm, falling above 200 nm — attributed to competition between surface curvature and specific surface area, and cross-checked against transition-state theory and a double-film mass-transfer treatment. Single salt system under controlled UV; the extrapolation to ambient mixed-composition marine aerosol is not demonstrated. Methodologically this is the interesting part: an optical single-particle technique that yields kinetics without collection, at sizes below what most PM instrumentation resolves at all.

doi: 10.5194/acp-26-12183-2026

Exposure assessment & modelling

5 records

Cui et al. 2026 Environ Pollut*Daily 1 km satellite-based PM_{2.5} exposure estimates in Bangladesh, 2020–2025*

Two-stage machine learning over one of the most monitoring-sparse high-burden regions in the world. Stage one reconstructs MAIAC aerosol optical depth, of which **more than 60% of potential retrievals are missing**, with LightGBM; stage two feeds the gap-filled AOD plus meteorological, composition and geospatial predictors into an attention-based TabNet to estimate ground PM_{2.5}. Independent test performance $R^2 = 0.95$, **RMSE 17.41 $\mu\text{g}/\text{m}^3$, MAE 9.38**, against a national annual mean of **68–75 $\mu\text{g}/\text{m}^3$** and winter peaks above 130. The limitation is structural: with almost no ground reference network, the held-out set is small and drawn from the same sparse monitors used in training, so $R^2 = 0.95$ describes interpolation skill more than transferability, and the DHS linkage is division-level and ecological.

doi: 10.1016/j.envpol.2026.129051

Hopke et al. 2026 Environ Monit Assess*A PM_{2.5} magnitude-rank index for summarizing annual concentration-frequency structure*

Introduces the PM_{2.5} Magnitude–Rank Index — the largest integer k such that at least k days in a year have daily mean $\text{PM}_{2.5} \geq k \mu\text{g}/\text{m}^3$ — as a threshold-free joint summary of magnitude and frequency. Applied to **918 qualifying MSA-years across 96 US metropolitan areas, 2013–2022**, PMRI ranges 6–38 (median 16) and correlates with the annual mean at Spearman $\rho = 0.83$, the annual 98th percentile at 0.95, and days $\geq 15 \mu\text{g}/\text{m}^3$ at 0.96. That last figure is the honest problem with the metric: at $\rho = 0.96$ against an existing exceedance count it adds very little information, so its case rests on being policy-threshold-free and intuitively readable rather than on discriminating power. No health-outcome validation is attempted.

doi: 10.1007/s10661-026-15858-1

Reyna-Bello et al. 2026 Environ Monit Assess*Seasonal variability of PM_{2.5}-bound metals and meteorology in an urban-industrial area of northern Mexico*

MiniVol low-volume sampling across four seasons in Saltillo during 2023, with ten metals (Al, Cd, Cu, Fe, K, Mn, Na, Ni, Pb, Zn) by flame atomic absorption and PCA against temperature, humidity, wind, precipitation and pressure. Mass is strongly cold-season weighted — **152.9 $\pm$ 23.6 $\mu\text{g}/\text{m}^3$ in autumn and 115.3 $\pm$ 37.2 in winter against 40.7 $\pm$ 6.8 in spring** — consistent with suppressed dispersion, while several metals invert that pattern and peak in spring and summer. The reported metal concentrations are implausibly high for PM_{2.5}-bound species (Cd at 3.13 $\mu\text{g}/\text{m}^3$ is orders of magnitude above typical urban values), which points to a units or digestion-blank problem; treat the metal magnitudes as unverified and the seasonal contrast as the usable result.

doi: 10.1007/s10661-026-15856-3

Leibensperger et al. 2026 Atmos Chem Phys*Sea surface warming suppresses primary sea spray aerosol number production*

Controlled wave-breaking experiments in the Scripps Ocean-Atmosphere Research Simulator, which reproduces wind, waves and sea surface temperature together. Raising SST from 2 to 23 °C **suppressed subsurface bubble concentration by 1.5 $\times$, sea spray aerosol number by 3 $\times$ and SSA number flux by 3 $\times$** , with a non-monotonic SST dependence in the supermicron fraction. This is a single simulator with one seawater matrix and no biological aging, so absolute fluxes are not transferable — the value is the sign and rough magnitude of the SST dependence, which the field has not agreed on. Relevance here is indirect but real: marine background number concentration is the baseline against which coastal low-cost network anomalies are read.

doi: 10.5194/acp-26-12211-2026

Agpaoa et al. 2026 Environ Monit Assess*Occurrence, temporal variation and meteorological drivers of airborne fungal spores in Laoag City, northern Philippines*

First aeromycological record for a monsoon-influenced coastal site in northern Luzon: 216 sampling days, March to November 2022, cumulative index **190,096 spores m^{-3}** , dominated by *Cladosporium* and Fusaroid-like types. Wet-season sampling carries most of the cumulative load but the highest daily concentrations fall in the dry season, and diurnal peaks sit at 00:00–06:00. Positive associations with relative humidity and rainfall, negative with temperature, and no significant wind association. Single site, single year, morphology-based identification only, and no PM co-measurement — so the bioaerosol fraction cannot be placed within a mass-based PM budget. Included because overnight spore peaks are a documented confounder for optical particle counters in humid climates.

doi: 10.1007/s10661-026-15814-z

Burden, policy & mitigation

4 records

Zhang et al. 2026g Nat Commun*Health and economic impacts of improved air quality from transitioning global shipping energy to hydrogen*

Integrated emission–air quality–health framework over feasible hydrogen-transition scenarios for global shipping. Full adoption averts approximately **129,500 (95% CI 68,300–202,200) premature deaths annually**, valued at **US\$306 bn (146–488)**, with India, Japan and the United States taking the largest economic benefit. The number that matters for a PM watch is the split: **75% of the averted mortality is ozone, only 25% PM_{2.5}**, with the O₃ gains concentrated in Southern Asia and the PM_{2.5} gains in Eastern Asia. The confidence interval spans a factor of three and propagates concentration–response uncertainty only, not scenario or inventory uncertainty, so the true spread is wider than stated.

doi: 10.1038/s41467-026-76134-x

Nilsson et al. 2026 Environ Epidemiol*Urban environment, health and equity: symptoms and annoyances related to air pollution, traffic noise and greenness*

46,282 adults aged 18–84 in Stockholm County responding to the 2023 Swedish National Environmental Health Survey, with geocoded residential PM_{2.5}, PM₁₀, NO₂, road/rail/aircraft noise and 300 m NDVI, dichotomised and stratified by socioeconomic status and building type. Apartment residents are more exposed to pollution and noise and have less greenness; SES-stratified patterns are exposure-specific with no consistent overall gradient. The strong associations are with *perception* — NO₂ with poor perceived outdoor air quality (**OR 3.22; 95% CI 2.75–3.79**), road noise with high annoyance (OR 4.59; 4.02–5.26). Both exposure and outcome are cross-sectional and the outcome is self-reported, so these are perception–exposure concordance estimates, not health effects, and the odds ratios should not be read as such.

doi: 10.1097/EE9.0000000000000516

Sirikul et al. 2026 J Clim Chang Health*Knowledge, attitudes and practices of caregivers regarding PM_{2.5} exposure among children in northern Thailand*

Cross-sectional KAP survey of caregivers recruited at Chiang Mai Child Development Centres. Knowledge is moderate (10.44±3.80 of 22), attitudes positive (30.74±3.57 of 40), practices inconsistent (17.33±5.22 of 32). The specific gaps are the useful part: **97.0% recognise the health risk to children, but only 41.8% know surgical masks are inadequate for PM_{2.5} and only 9.7% can correctly interpret an AQI value**; air purifier ownership is 15.3% and monitoring-app use 28.2%, both with a clear education and income gradient. The abstract does not report the sample size, response rate or instrument validation, and centre-based recruitment excludes non-attending households, so absolute prevalences are not generalisable. The AQI-literacy figure is the one that should worry anyone designing a public-facing sensor dashboard.

doi: 10.1016/j.joclim.2026.100734

Sharma et al. 2026b Health Promot J Austr*Geospatial linkage of environmental exposures, socioeconomic profile and health outcomes among women of reproductive age, Victoria*

Postcode-level linkage of 2021 Australian Census sociodemographic and health variables to PM_{2.5}, NO₂, CO, SO₂, formaldehyde, ozone, land surface temperature and greenness, with Moran's I, Wagstaff concentration indices and geospatial regression. Non-communicable disease burden in women of reproductive age is strongly socioeconomically patterned (**323.0 vs. 227.3 per 1000** in most- versus least-disadvantaged postcodes; steepest for diabetes, Wagstaff –0.16), **but environmental exposures show only weak associations with outcomes** despite clear spatial clustering. Inner Melbourne has lower NCD burden despite elevated NO₂ and PM_{2.5}. This is ecological, cross-sectional and confounded by the same SES gradient it measures; the null environmental finding is uninformative rather than reassuring.

doi: 10.1002/hpja.70241

Neuro / mental health

2 records

Wang et al. 2026e Sleep Med*Speciated fine particulate matter and self-reported sleep time in middle-aged and older Chinese adults*

19,505 CHARLS participants aged 45+, 2011–2018, with 6-month and 2-year $PM_{2.5}$ and five constituents — BC, organic matter, SO_4^{2-} , NO_3^- , NH_4^+ — from the Tracking Air Pollution in China product, against total sleep time (nocturnal plus napping) via linear mixed models and quantile g-computation. The secondary inorganic ions NO_3^- , SO_4^{2-} and NH_4^+ show the most consistent associations, and the direction is the surprise: highest versus lowest quartile of intermediate-term exposure **increases** total sleep time ($PM_{2.5} + 0.16$ h and comparable for the constituents). Self-reported duration, no actigraphy, and an increase in sleep duration is not obviously a harm — it may index fatigue, reduced outdoor activity on high-pollution days, or reverse behavioural adaptation. Mechanism is not addressed and the finding should be treated as hypothesis-generating.

doi: 10.1016/j.sleep.2026.109229

Chen et al. 2026i J Occup Environ Med*Gender and season disparities in the influence of short-term $PM_{2.5}$ exposure on emergency room visits for mood disorders in Taiwan*

Time-stratified case-crossover with distributed lag non-linear models over **2,172 emergency visits** for bipolar or depressive disorder in the Chang Gung Research Database, 2010–2020, linked to Ministry of Environment monitoring. Associations appear **exclusively in the hot season and predominantly in women**, with delayed effects at lags 3–5 and cumulative effects over lags 0–1 and 0–3 to 0–5. No effect estimates, confidence intervals or exposure contrasts appear in the abstract, which makes the result unquantifiable as reported. With 2,172 events split by sex and season the subgroup cells are small, and a hot-season-only, women-only finding from a design with no pre-registered subgroup plan is the classic shape of a multiplicity artefact. Carried at tier B on design quality, not on the strength of the reported signal.

doi: 10.1097/JOM.0000000000003848

Cardiovascular & metabolic

1 record

Li et al. 2026e Environ Sci Technol Lett

Sulfate-dominated PM_{2.5} constituents are associated with increased acute myocardial infarction mortality: a US nationwide difference-in-differences study

County-level US AMI mortality against PM_{2.5} and its EC, OC, SO₄²⁻ and NO₃⁻ constituents, analysed with a causal mixture method inside a difference-in-differences design. Total mass: **RR 1.019 (95% CI 1.013–1.025) per IQR**. Constituent mixture: **RR 1.116 (1.101–1.130) per one-quartile increase**, with SO₄²⁻ **carrying weight 0.762**. Regionally, an eastern county sees **1.545 (0.656–2.437) excess AMI deaths per 100,000 per year** relative to the west per quartile of mixture, SO₄²⁻-driven in the east (0.707) and OC-driven in the west (0.485). Exposure is county-level and the outcome is mortality rather than incidence, so this is ecological in structure — the DiD design controls time-invariant county confounding but not time-varying composition-correlated confounders such as regional industrial decline.

doi: 10.1021/acs.estlett.6c00288

Mechanistic toxicology

1 record

Van et al. 2026 In Vivo

MASLD exacerbates chronic low-dose PM_{2.5}-induced lung injury, inflammation and fibrosis

C57BL/6J mice on a Western or normal diet for 28 weeks with concurrent intratracheal diesel particulate matter at **0.5 mg/kg three times weekly** (stated as environmentally relevant to ~50 µg/m³), assessed by histology, Masson's trichrome, Luminex multiplex on bronchoalveolar lavage and lung-tissue ELISA/qPCR. The Western diet induced the MASLD phenotype; PM_{2.5} did not worsen the hepatic phenotype, but **the combination markedly exacerbated pulmonary injury and fibrosis relative to PM_{2.5} alone**. The dose-equivalence claim is the weak point: bolus intratracheal instillation of diesel PM delivers a deposition profile nothing like chronic inhalation at 50 µg/m³, so the “environmentally relevant” framing should not be taken at face value. The interaction direction is the transferable result.

doi: 10.21873/invivo.14428

Occupational & indoor

1 record

Ospalek et al. 2026 Atmosphere*Air quality assessment and potential inhalation exposure across equestrian facilities*

Nine measurements at 1.5 m in a stable, an indoor riding arena and an outdoor riding arena at a Slovak university equestrian centre, covering PM₁, PM_{2.5}, PM₁₀, respirable and total dust plus gaseous pollutants, with four exposure scenarios combining 18–20 h stabled with 4–6 h ridden. The **outdoor arena carried the highest concentrations of every PM fraction and the indoor arena the lowest** ($p < 0.05$) — the opposite of the expected enclosure effect, and the paper's most interesting result; ammonia was higher in the stable, as expected. Summer measurements only, one facility, nine measurements total, and the exposure scenarios are constructed rather than observed, so the inhaled-dose estimates are illustrative. Relevant here as a template for microenvironment-resolved dose reconstruction from sparse fixed-point sampling.

doi: 10.3390/atmos17090840

Other clinical endpoints

1 record

Wu et al. 2026d J Am Coll Radiol*Short-term environmental exposures and emergency department imaging utilization*

Time-stratified case-crossover over **1,946,465 imaging studies** at five Toronto academic hospitals, 2013–2023, with conditional Poisson models against PM_{2.5}, NO₂, temperature, wind, rainfall and snowfall, modelling both absolute volume and per-visit rate (log daily visits as offset). Higher PM_{2.5} associates with higher imaging volume, **IRR 1.020 (95% CI 1.017–1.024) per 10 µg/m³**, alongside NO₂ 1.017 (1.014–1.020) per 10 ppb and temperature 1.040 per 10 °C; wind, rain and snow are protective, snowfall most strongly (0.959 per 10 mm). The volume effect is almost certainly mediated by visit counts rather than by per-visit imaging intensity, and the exposures are collinear with each other and with season — so this is a health-system demand signal, not evidence of a PM-specific clinical mechanism. Its value is operational: PM_{2.5} forecasts carry usable information for ED resource planning.

doi: 10.1016/j.jacr.2026.08.014

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Chen et al.	Gender and season disparities in the influence of short-term PM _{2.5} exposure on emergency room visits for mood disorders in Taiwan	<i>J Occup Environ Med</i>	Time-stratified case-crossover + DLNM	Daily PM _{2.5} , lags 0-5	Taiwan
Wang et al.	Speciated fine particulate matter and self-reported sleep time in middle-aged and older Chinese adults	<i>Sleep Med</i>	Prospective cohort (CHARLS)	6-month and 2-year PM _{2.5} plus BC, OM, SO ₄ , NO ₃ , NH ₄ (TAP dataset)	China
Cardiovascular & metabolic					
Li et al.	Sulfate-dominated PM _{2.5} constituents are associated with increased acute myocardial infarction mortality: a US nationwide difference-in-differences study	<i>Environ Sci Technol Lett</i>	Quasi-experimental (difference-in-differences)	PM _{2.5} mass and EC / OC / SO ₄ / NO ₃ constituent mixture, per IQR	United States
Mechanistic toxicology					
Van et al.	MASLD exacerbates chronic low-dose PM _{2.5} -induced lung injury, inflammation and fibrosis	<i>In Vivo</i>	Animal model (chronic instillation)	Diesel PM 0.5 mg/kg intratracheal 3x/week (50 µg/m ³ equivalent)	Taiwan
Occupational & indoor					
Ospalek et al.	Air quality assessment and potential inhalation exposure across equestrian facilities: a case study	<i>Atmosphere</i>	Cross-sectional environmental survey	PM ₁ , PM _{2.5} , PM ₁₀ , respirable and total dust at 1.5 m; NH ₃	Slovakia
Sensing, forecasting & instrumentation					
Marcon et al.	Citizen science in air pollution monitoring: methodological approaches, effectiveness, and integration with health research	<i>Curr Environ Health Rep</i>	Critical review	Low-cost sensor networks, wearables, AI-assisted calibration	Global / multi-region
Wongsasuluk et al.	PM _{2.5} exposure and respiratory health risk for vulnerable populations in an urban underpass: a predictive MLR approach, Chiang Mai	<i>Environ Sci Pollut Res Int</i>	Network deployment + statistical model	PM _{2.5} and PM ₁₀ from five DustBoy low-cost stations, 10-min cadence	Thailand
Xiong et al.	In situ real-time determination of SO ₂ photochemical oxidation in nanoscale sea salt aerosols based on dark-field microscopy	<i>Atmos Chem Phys</i>	Laboratory instrument development	Single-nanoparticle hygroscopic growth factor; first-order rate constant	China
Zhang et al.	Contrasting nighttime heterogeneous and daytime photochemical aging drive the optical evolution of black carbon	<i>Atmos Chem Phys</i>	Field campaign, single-particle instrumentation	rBC mass, core MMD, coating-to-core mass ratio, MAC at 532 nm (SP2 + SPAMS)	China
Exposure assessment & modelling					
Agpaoa et al.	Occurrence, temporal variation and meteorological drivers of airborne fungal spores in Laoag City, northern Philippines	<i>Environ Monit Assess</i>	Bioaerosol monitoring campaign	Volumetric spore counts (spores/m ³), six spore types	Philippines
Cui et al.	Daily 1 km satellite-based PM _{2.5} exposure estimates in Bangladesh, 2020-2025	<i>Environ Pollut</i>	Satellite-surface fusion, machine learning	Gap-filled MA-IAC AOD -> ground PM _{2.5} (LightGBM + TabNet)	Bangladesh
Hopke et al.	A PM _{2.5} magnitude-rank index for summarizing annual concentration-frequency structure	<i>Environ Monit Assess</i>	Metric development + retrospective application	PMRI: largest k with >= k days at >= k µg/m ³	United States
Leibensperger et al.	Sea surface warming suppresses primary sea spray aerosol number production	<i>Atmos Chem Phys</i>	Controlled wave-channel experiment	Bubble and SSA number concentration and emission flux, 0.008-20 µm	Global / multi-region
Reyna-Bello et al.	Seasonal variability of PM _{2.5} -bound metals and meteorology in an urban-industrial area of northern Mexico: a PCA assessment	<i>Environ Monit Assess</i>	Filter sampling + composition	MiniVol low-volume PM _{2.5} mass; 10 metals by FAAS	Mexico
Burden, policy & mitigation					

First author	Focus	Journal	Design	Metric	Region
Nilsson et al.	Urban environment, health and equity: symptoms and annoyances related to air pollution, traffic noise and greenness	<i>Environ Epidemiol</i>	Cross-sectional population survey	Dichotomised residential PM _{2.5} , PM ₁₀ , NO ₂ at geocoded address	Sweden
Sharma et al.	Geospatial linkage of environmental exposures, socioeconomic profile and health outcomes among women of reproductive age, Victoria	<i>Health Promot J Austr</i>	Ecological geospatial linkage	Postcode-level PM _{2.5} , NO ₂ , CO, SO ₂ , HCHO, O ₃ , LST, greenness	Australia
Sirikul et al.	Knowledge, attitudes and practices of caregivers regarding PM _{2.5} exposure among children in northern Thailand	<i>J Clim Chang Health</i>	Cross-sectional survey	None measured; AQI literacy and protective behaviour	Thailand
Zhang et al.	Health and economic impacts of improved air quality from transitioning global shipping energy to hydrogen	<i>Nat Commun</i>	Integrated emission-air quality-health modelling	Modelled PM _{2.5} and O ₃ concentration change under hydrogen scenarios	Global / multi-region
Other clinical endpoints					
Wu et al.	Short-term environmental exposures and emergency department imaging utilization	<i>J Am Coll Radiol</i>	Time-stratified case-crossover	Daily PM _{2.5} , per 10 µg/m ³	Canada

Provenance and method

Entry window **2026-08-28 to 2026-08-28**, a single day of deposits. **Sources.** PubMed E-utilities on [Date - Entry], run as two separate queries (health axis 17, instrumentation axis 2, one record shared). Europe PMC CREATION_DATE returned 0 for this date. Crossref by-ISSN across the tracked non-PubMed-indexed titles returned 57 deposits. OpenAlex returned HTTP 429 and remains broken for entry-date windowing; arXiv contributed nothing dated in window. **Screening.** 76 raw records, **73 unique** after PMID/DOI collapse (one PubMed record returned on both axes, one Crossref deposit duplicated within the feed, and the Bangladesh paper arrived through both PubMed and Crossref), **0 already in seen.json**, **19 retained and 54 rejected** with reason logged to `state/rejected.jsonl`. Rejections break down as 43 out of scope (non-particle atmospheric chemistry, water and soil contamination, NDT and metrology deposits from *Meas Sci Technol*), 8 in scope by title but with **no retrievable abstract** — carried to the retry list rather than summarised — and 3 editorial front matter (two retraction notices, one corrigendum). **Effect estimates.** Four estimates from two studies; the figure caption states why none of the four pairs is fully comparable. **Pipeline changes made this run.** Seventeen design labels and *Slovakia* were added to `plots.py` in the same edit that wrote them onto records, per the 22 August guard. **Labels.** Subtopic, design, particle-metric and geography are single-label by dominant contribution — a judgement, not a controlled vocabulary. All 19 DOIs passed `check_dois.py` (0 fail, 0 warn) before build.